EE352 Automatic Control

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Time Domain Analysis

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2. Typical test signals for time response of continuous data systems
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Time response of continuous data systems

- When you are in the **time domain**, time is the **independent variable** in all control systems
- Make sense to evaluate the **States** and the **Output(s)** with respect to **Time**
- This is called **Time Response**
- It is one of the main criteria in **Performance Evaluation** of a Control System
- Consists of two parts
  - Transient Response $y_t(t)$
  - Stead State Response $y_{ss}(t)$
- Time Response $\mathrm{y}(t) = y_t(t) + y_{ss}(t), \lim_{t\to\infty} y_t(t) = 0$

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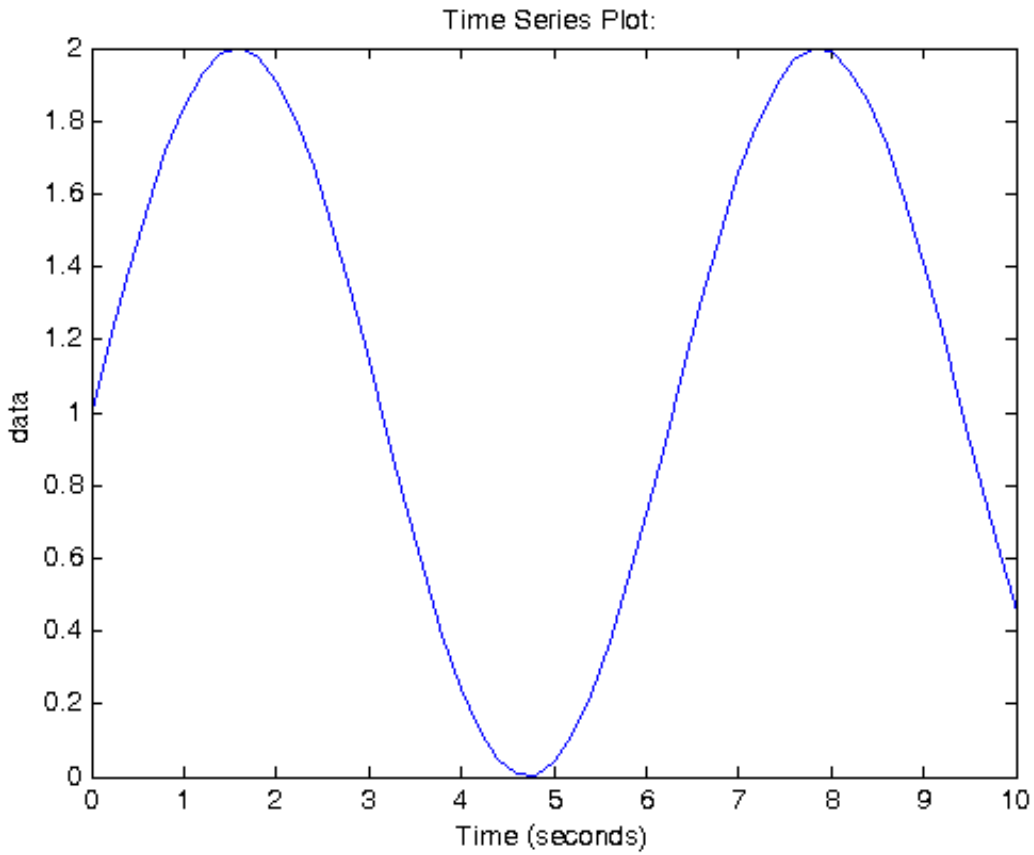
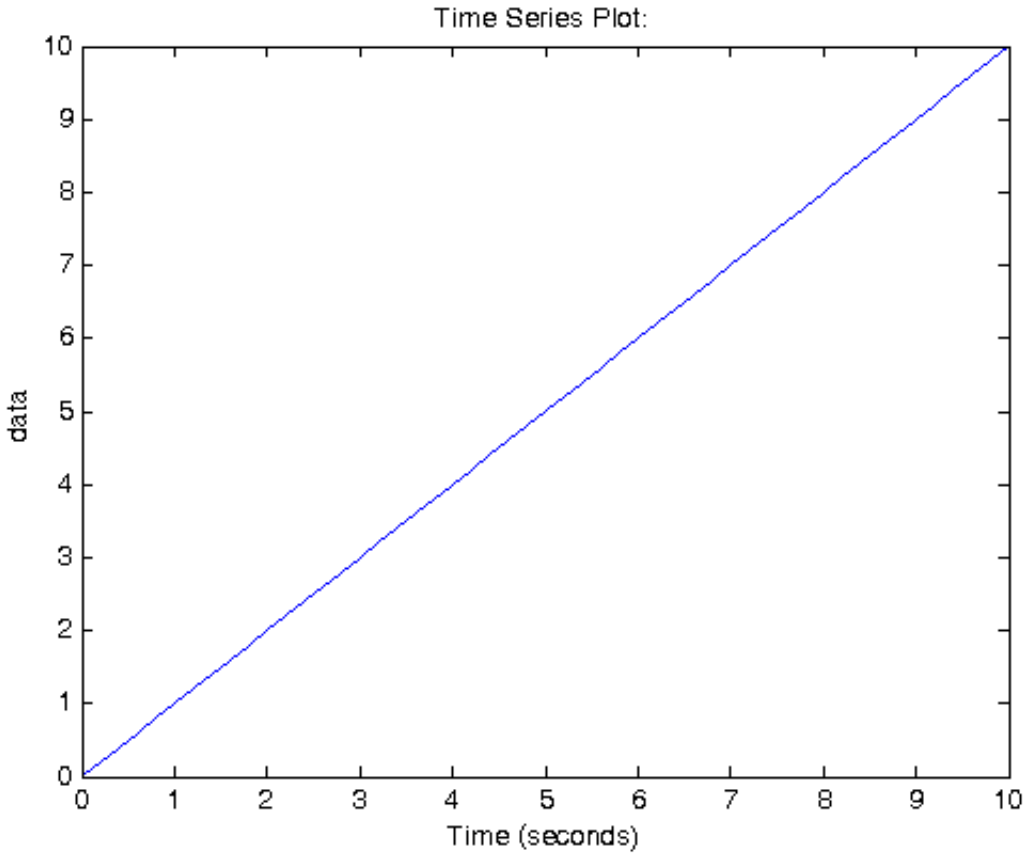
Time varying Steady State responses

Sinusoidal output



Ramp output



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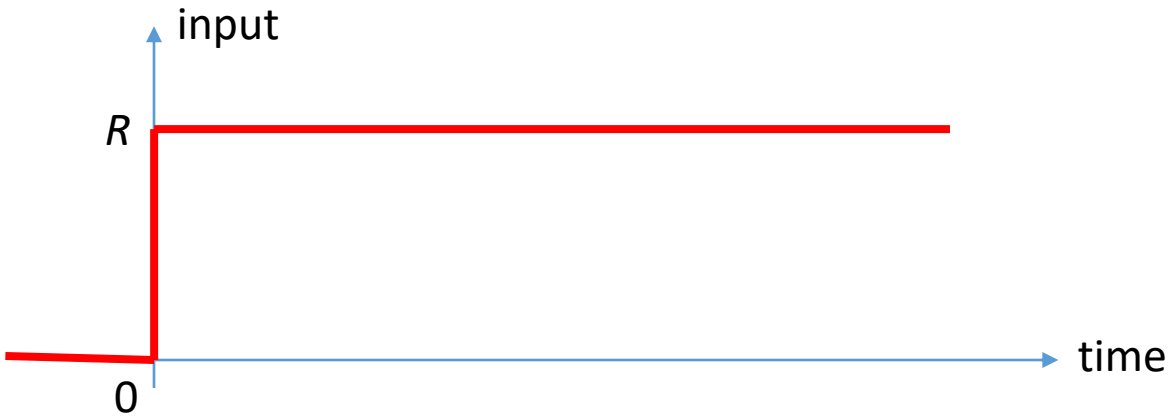
Steady State Error

- **IF** Steady State Response $\neq$ Desired Output,
- **THEN Steady State Error** $\neq 0$
- **Steady State Error** is a typical parameter in the design specification

Typical test signals for time response of continuous data systems

- Input signals to many practical control systems are not known in advance
- For analysis and design of control systems, it is necessary to assume some input signals which
  - Are not far from reality
  - Demonstrates the required performance
- Design specifications are made with respect the responses of those input signals
- Time domain specifications are made with respect to
  - Step function input and / or
  - Ramp function input and / or
  - Parabolic function input
- Frequency domain specifications are made with respect to a sinusoidal input

Step function input



Mathematical representation

$$r(t) = \begin{cases} R & t \geq 0 \\ 0 & t < 0 \end{cases}$$

where R is real constant

OR

$$r(t) = Ru_s(t)$$

where $u_s(t)$ is a unit step

Description

- The step response reveals how quick the control system will respond to a sudden change in the input.
- The step input contains a wide band of frequencies in the spectrum. Hence it is equivalent to applying a linear combination of sinusoidal signals of a wide band of frequencies

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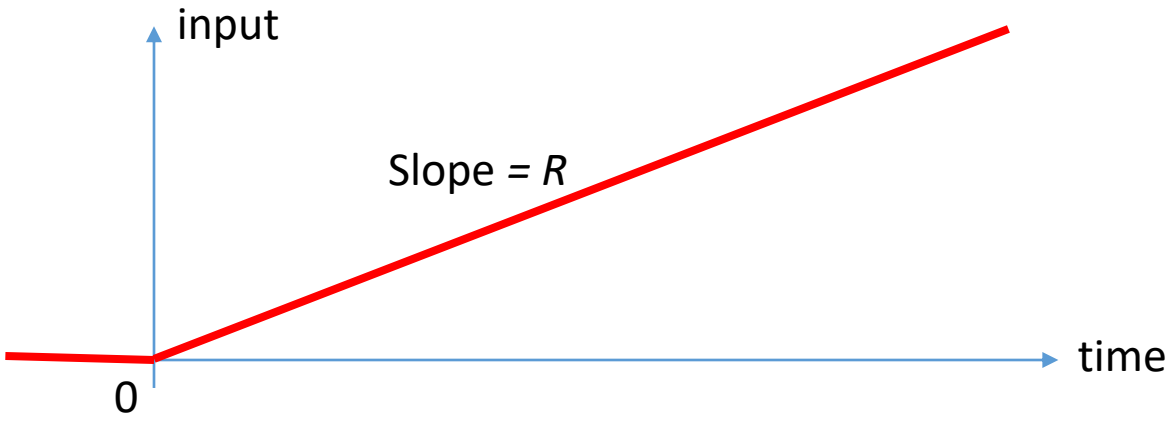
Ramp function input



Mathematical representation

$$r(t) = \begin{cases} Rt & t \geq 0 \\ 0 & t < 0 \end{cases}$$

where *R* is real constant

OR

$$r(t) = Rtu_s(t)$$

where $u_s(t)$ is a unit step

Description

- The ramp input may increase or decrease at a constant rate.
- The ramp input allows to test how the control system would respond to an input signal that changes linearly with time.

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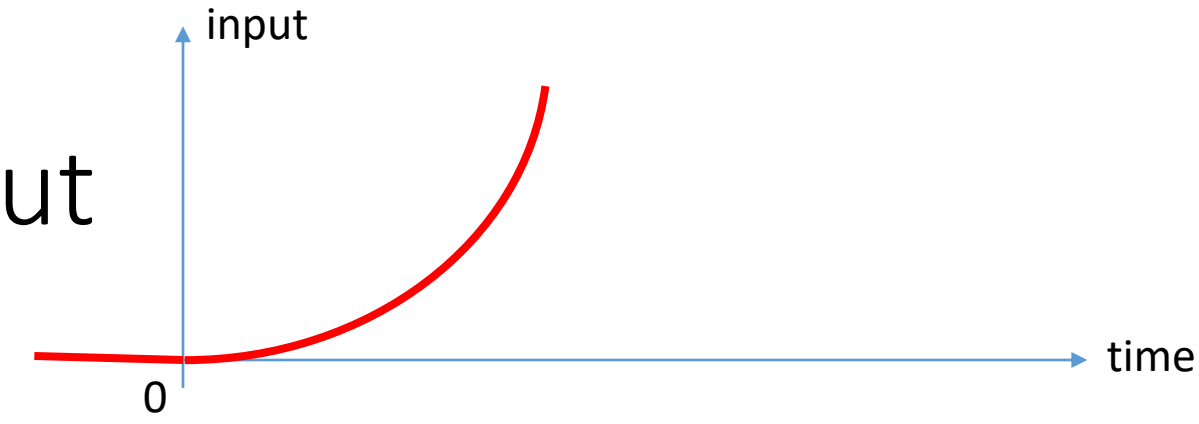
Parabolic function input



Mathematical representation

$$r(t) = \begin{cases} R\dfrac{t^2}{2} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

where *R* is real constant

OR

$$r(t) = R\frac{t^2}{2}u_s(t)$$

where $u_s(t)$ is a unit step

Description

- The ramp input may increase or decrease quadratically.
- $\frac{1}{2}$ is there because $\mathrm{R}(s) = \frac{R}{s^3}$
- The parabolic input allows to test how the control system would respond to an input signal that quadratically changes with time.

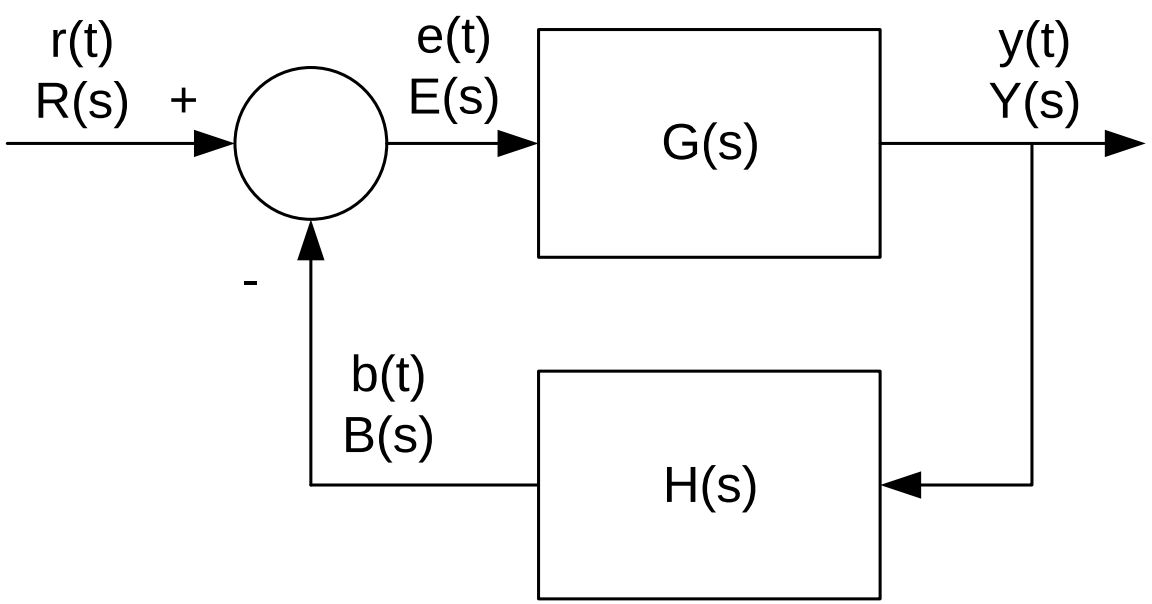
Following reference inputs

- Control system output is generally supposed to follow a specific reference signal accurately.
- Difference between the **Output** and the **Reference** in the steady state is the **Steady State Error (SSE)**
- Due to imperfections in the control system components, SSE exists in most systems
- One of the design objectives will be to keep the SSE minimum or below a certain threshold
- In general SSE of LTI systems depend on the type of input and the '**Type**' of the system
- '**Type**' of the system is the number of poles at the origin

Definition of SSE with respect to system configuration

- Reference is the signal that the output $y(t)$ has to track.
- It may not be $r(t)$ in the figure.
- In unity feedback systems, $H(s) = 1$ and it is $r(t)$
- For unity feedback systems $e(t) = r(t) - y(t)$
- In either case, the SSE is defined as $e_{ss} = \lim_{t \to \infty} e(t)$

$$e(t) = reference - y(t)$$



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SSE contd.,

- When $H(s) \neq 1, u(t)$ may or may not be the error depending on the form and the purpose of $H(s)$.
- Assume that the objective of the control system is to have the output $y(t)$ track the input $r(t)$ as closely as possible.

CASE I: $G(s) = \frac{1}{s^2(s+12)}; H(s) = 1; CE \equiv s^3 + 12s^2 + 1 = 0$

Poles are at $s = -12.0069\ and\ s = 0.0035 \pm 0.2886j$

Hence the system is unstable and no way to define the error

CASE II: $G(s) = \frac{1}{s^2(s+12)}; H(s) = \frac{5(s+1)}{(s+5)}; CE \equiv s^4 + 17s^3 + 60s^2 + 5s + 5 = 0$

Poles are at $s = -12.0539, s = -4.8856\ and\ s = -0.0302 \pm 0.2898j$

Hence the error may be defined as $e(t) = r(t) - y(t)$

SSE contd.,

- If the above setup is converted into a velocity control system, in which case a step input is used to control the system output, which contain a ramp at steady state, the transfer function may be of the form
- **CASE III**: $G(s) = \frac{1}{s^2(s+12)}; H(s) = K_v s$
- $H(s)$ is the transfer function of the tachometer and K_v is the tachometer constant.
- In this case, the error has to be defined as

$$e(t) = reference\ - y(t)$$

- The reference signal is the **desired velocity** and **NOT** $\boldsymbol{r(t)}$
- Also, $y(t)$ and $r(t)$ are not of the same dimensions. Hence no meaning of defining the error as $e(t) = r(t) - y(t)$

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SSE contd.,

- For example, if $K_v = 10\ V/rads^{-1}$ then for a unit step of $1\ V$, the desired velocity at steady state is ${}^{1V}/_{10\ V/rads^{-1}} = 0.1rads^{-1}$
- With $G(s) = \frac{1}{s^2(s+12)}$; $H(s) = 10s$ the closed loop transfer function is

$$M(s) = \frac{Y(s)}{R(s)} = \frac{1}{s(s^2+12s+10)}.$$

- With unit step input $R(s) = \frac{1}{s}$, the output is

$$y(t) = \mathcal{L}^{-1}\left(\frac{1}{s^2(s^2+12s+10)}\right)$$
$$= 0.1t - 0.12 - 0.000796e^{-11.1t} + 0.1208e^{-0.901t}$$

- The steady state output $y_{ss} = 0.1t - 0.12$
- Since the desired velocity is $0.1t\ rads^{-1}$, the SSE is

$$e_{ss} = 0.1t - (0.1t - 0.12\ ) = 0.12$$

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‘Type’ of control systems

- With unity feedback $E(s) = \frac{1}{1+G(s)} R(s)$
- At steady state $e_{ss} = \lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) = \lim_{s\to 0} \frac{s}{1+G(s)} R(s)$
- Hence e_{ss} depends on the characteristics $G(s)$, specifically the number of poles at the origin.
- The number of poles at the origin is known as the **‘Type’** of the system
- Example1: $G(s) = \frac{K(1+0.5s)}{s(1+s)(1+2s)(1+s+s^2)}$ is a Type 1 system
- Example2: $G(s) = \frac{K(1+2s)}{s^3}$ is a Type 3 system
- Example3: $G(s) = \frac{K(1+5s)}{(1+s)(1+2s)}$ is a Type 0 system

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SSE of System with Step function input

- Step function input $R(s) = \frac{R}{s}$
- $e_{ss} = \lim_{s \to 0} \frac{s}{1+G(s)} R(s) = \lim_{s \to 0} \frac{R}{1+G(s)} = \frac{R}{1+\lim_{s \to 0} G(s)}$
- Defining the step error constant $K_p = \lim_{s \to 0} G(s)$
- $e_{ss} = \frac{R}{1+K_p}$
- For Type 0 system, $e_{ss} = \frac{R}{1+K_p}$ is a constant
- For Type 1 or higher system, $e_{ss} = 0$

SSE of System with Ramp function input

- Ramp function input $R(s) = \frac{R}{s^2}$
- $e_{ss} = \lim_{s \to 0} \frac{s}{1+G(s)} R(s) = \lim_{s \to 0} \frac{R}{s+sG(s)} = \frac{R}{\lim_{s \to 0} sG(s)}$
- Defining the ramp error constant $K_v = \lim_{s \to 0} sG(s)$
- $e_{ss} = \frac{R}{K_v}$
- For Type 0 system, $e_{ss} = \infty$
- For Type 1 system, $e_{ss} = \frac{R}{K_v}$ is a constant
- For Type 2 or higher system, $e_{ss} = 0$

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SSE of System with Parabolic function input

- Parabolic function input $R(s) = \frac{R}{s^3}$
- $e_{ss} = \lim_{s \to 0} \frac{s}{1+G(s)} R(s) = \lim_{s \to 0} \frac{R}{s^2 + s^2 G(s)} = \frac{R}{\lim_{s \to 0} s^2 G(s)}$
- Defining the parabolic error constant $K_a = \lim_{s \to 0} s^2 G(s)$
- $e_{ss} = \frac{R}{K_a}$
- For Type 0 and Type 1 systems, $e_{ss} = \infty$
- For Type 2 system, $e_{ss} = \frac{R}{K_a}$ is a constant
- For Type 3 or higher system, $e_{ss} = 0$

Example: Consider $G(s) = \dfrac{K(s+3.15)}{s(s+1.5)(s+0.5)}; H(s) = 1$

Input	Error constant	e_{ss}
Step	$K_p = \lim_{s\to 0} G(s) = \infty$	$\dfrac{R}{1+K_p} = 0$
Ramp	$K_v = \lim_{s\to 0} sG(s) = 4.2K$	$\dfrac{R}{K_v} = \dfrac{R}{4.2K}$
Parabolic	$K_a = \lim_{s\to 0} s^2 G(s)$	$\dfrac{R}{K_a} = \infty$

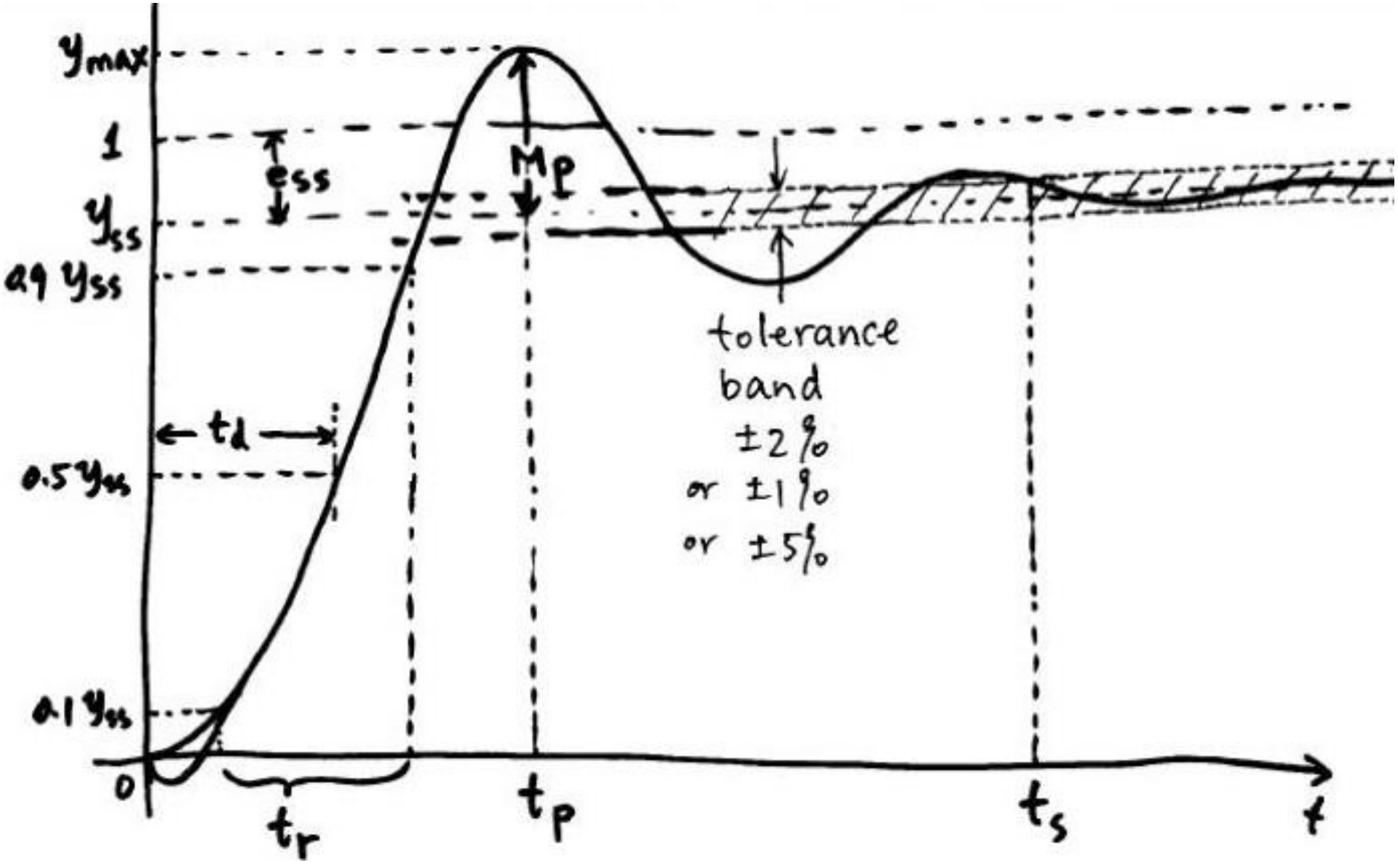
Unit step response and time domain specifications

- Time Response $\mathrm{y}(t) = y_t(t) + y_{ss}(t), \lim_{t\to\infty} y_t(t) = 0$
- However, the amplitude and time duration of $y_t(t)$ must be within limits acceptable according the performance expectations of the system
- For LTI systems, the characteristics of the transient response is often obtained from the Step Response.
- Hence, we look at the step response and define the performance criteria

Time domain performance criteria

- **Maximum overshoot**: $y_{max} - y_{ss}$
- **Maximum percentage overshoot**: $\left(\frac{y_{max} - y_{ss}}{y_{ss}}\right) \times 100\%$
- **Delay time (t_d)**: Time taken to reach 50% of the final value
- **Rise time (t_r)**: Time required for the step response to rise from 10% to 90% of the final value
- **Settling time (t_s)**: Time required for the step response to reach within $\pm 5\%$ of the final value
- These time domain specifications are easy to measure when the step response is well defined.
- Analytically these concepts are difficult to establish except for simple systems lower than 3rd order
- Many practical systems can be reduced to such systems using dominant pole analysis

Time domain performance criteria contd.,



Transient response of a Generic Second Order system

- Second order systems are rare in practice
- But their analysis help to form the basis for understanding of the analysis and design of higher order systems, especially the ones that can be approximated to second order systems
- Considering the Second Order forward transfer function
- $G(s) = \frac{Y(s)}{E(s)} = \frac{\omega_n^2}{s(s+2\zeta\omega_n)}$ where ω_n and ζ are constants.
- The unity feedback closed loop system is $\frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$
- This is called Generic (Prototype) Second Order system
- The Characteristic Equation (CE) of it is
- $CE \equiv s^2 + 2\zeta\omega_n s+\omega_n^2 = 0$

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Step response of a Generic Second Order system

- If a unit step is applied as the input, the output can be obtained in the s domain as

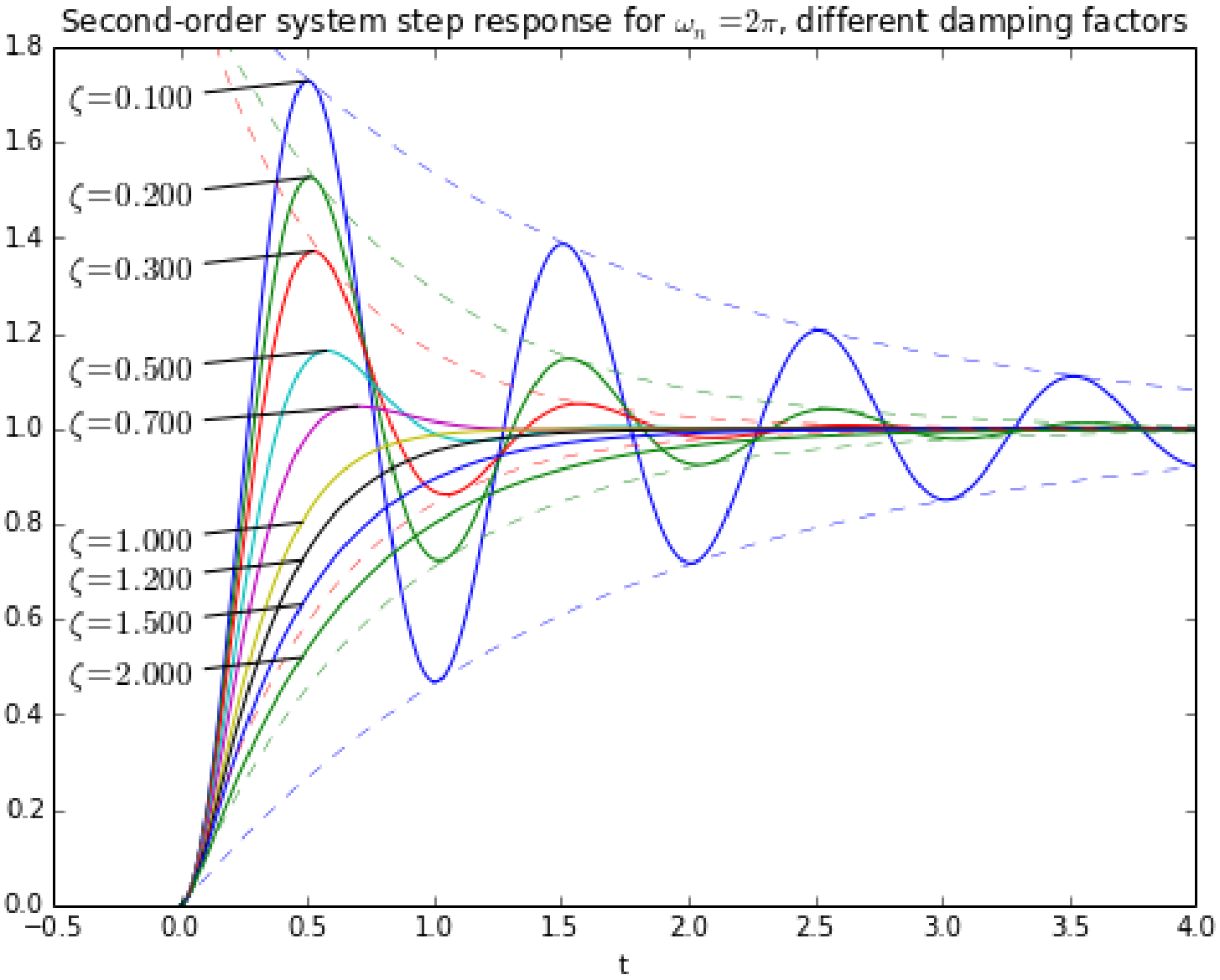
$$Y(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

- By taking inverse Laplace transforms

$$y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin\left(\omega_n\sqrt{1-\zeta^2}t + \cos^{-1}(\zeta)\right)$$

Generic Second Order systems contd.,

- As ζ decreases $(0 < \zeta < 1)$, the step response becomes oscillatory with higher overshoots
- When$\zeta \geq 1$, there are no overshoots.
- ω_n has some effect on the t_d, t_r and t_s, but does not affect the overshoot



Damping Factor

- The two roots of the Characteristic Equation $CE \equiv s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$ can be expressed as

$$s_1, s_2 = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -\alpha \pm j\omega$$

where $\alpha = \zeta\omega_n$ and $\omega = \omega_n\sqrt{1-\zeta^2}$

- We have $y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_n\sqrt{1-\zeta^2}t + \cos^{-1}(\zeta)\right)$, which is

$$y(t) = 1 - \frac{\omega_n e^{-\alpha t}}{\omega}\sin(\omega t + \cos^{-1}(\zeta))$$

- Hence α controls the rate of rise or the rate of decay of the step response.
- Therefore α is called ***Damping Factor*** or ***Damping Constant***.
- $\frac{1}{\alpha}$ is called the ***Time Constant*** of the system

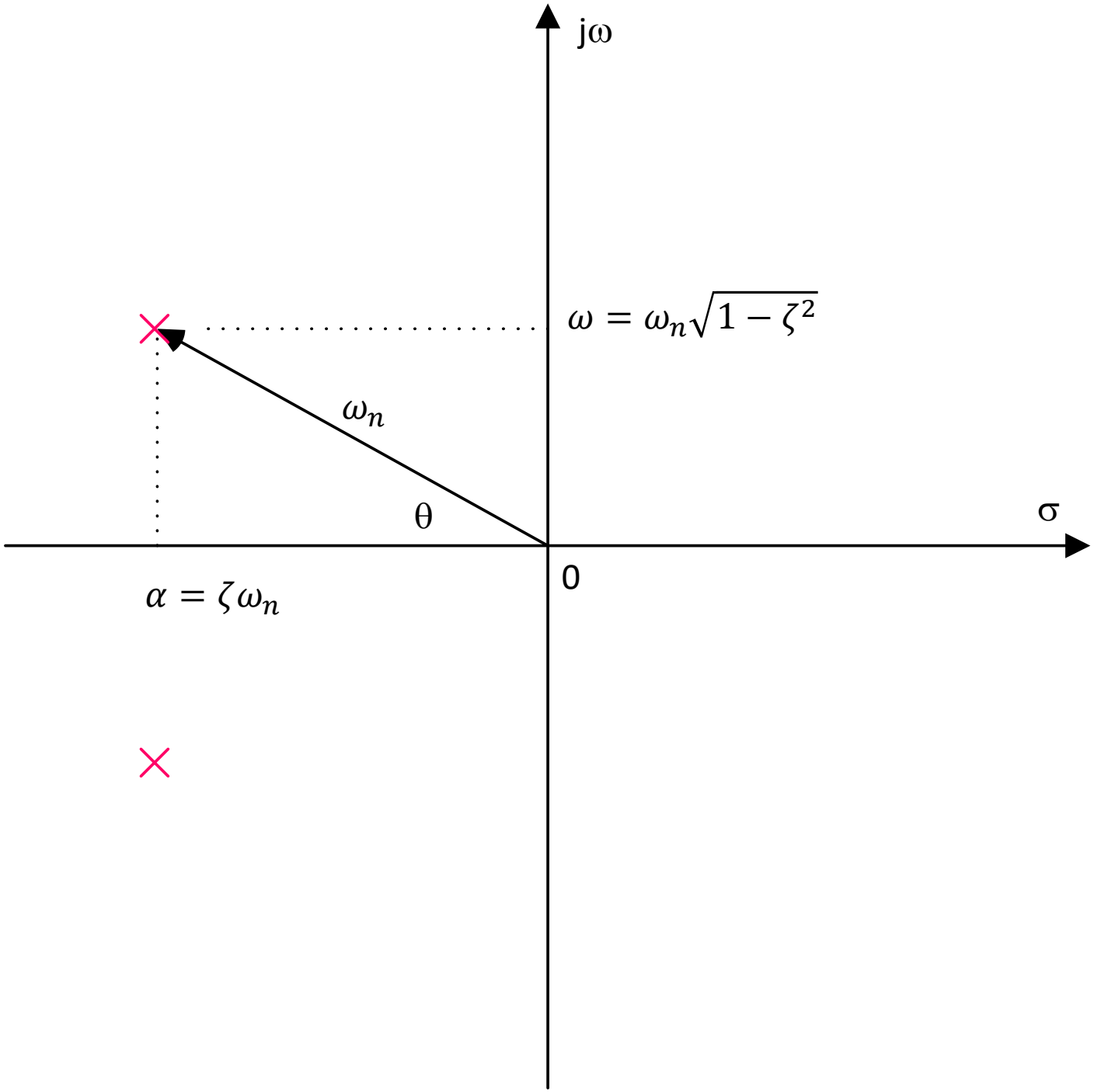
Damping Ratio

- When the two roots of $CE \equiv s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$ are real and equal, the system is Critically Damped. It occurs when $\zeta = 1$.
- When the system is Critically Damped, $\alpha = \omega_n$
- Hence, ζ becomes the Damping Ratio
- $\zeta = damping\ ratio = \dfrac{\alpha}{\omega_n} = \dfrac{actual\ damping\ factor}{damping\ factor\ at\ critical\ damping}$

Natural Undamped Frequency

- The parameter ω_n is defined as the ***Natural Undamped Frequency***
- When $\zeta = 0$, the system has poles at $s_1, s_2 = \pm j\omega_n$ and the system unit step response is purely sinusoidal.
- Therefore ω_n corresponds to the frequency of the undamped sinusoidal response.
- But when $0 < \zeta < 1$, the imaginary parts of the roots have the magnitude ω and $y(t)$ is not a periodic function.
- Hence ω is called Conditional Frequency or Damped Frequency.

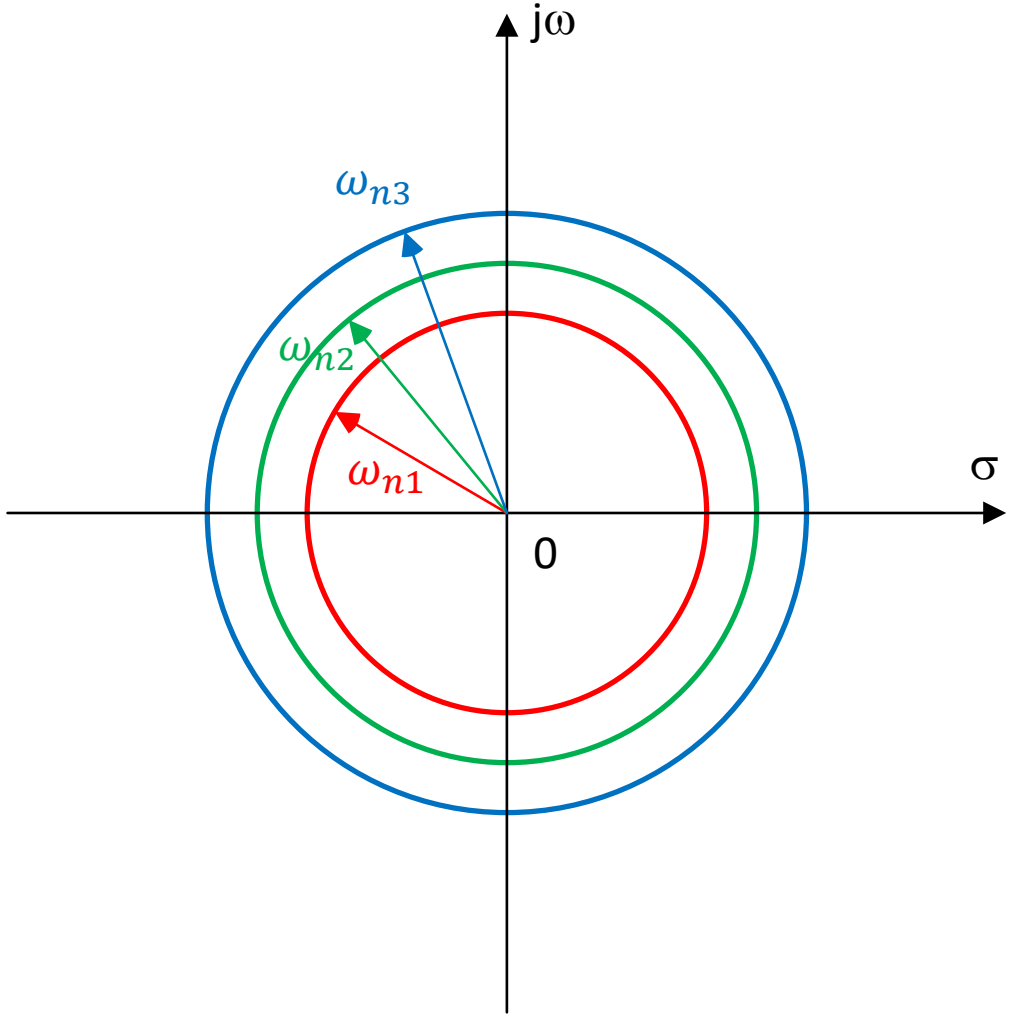
Second order system parameter representation



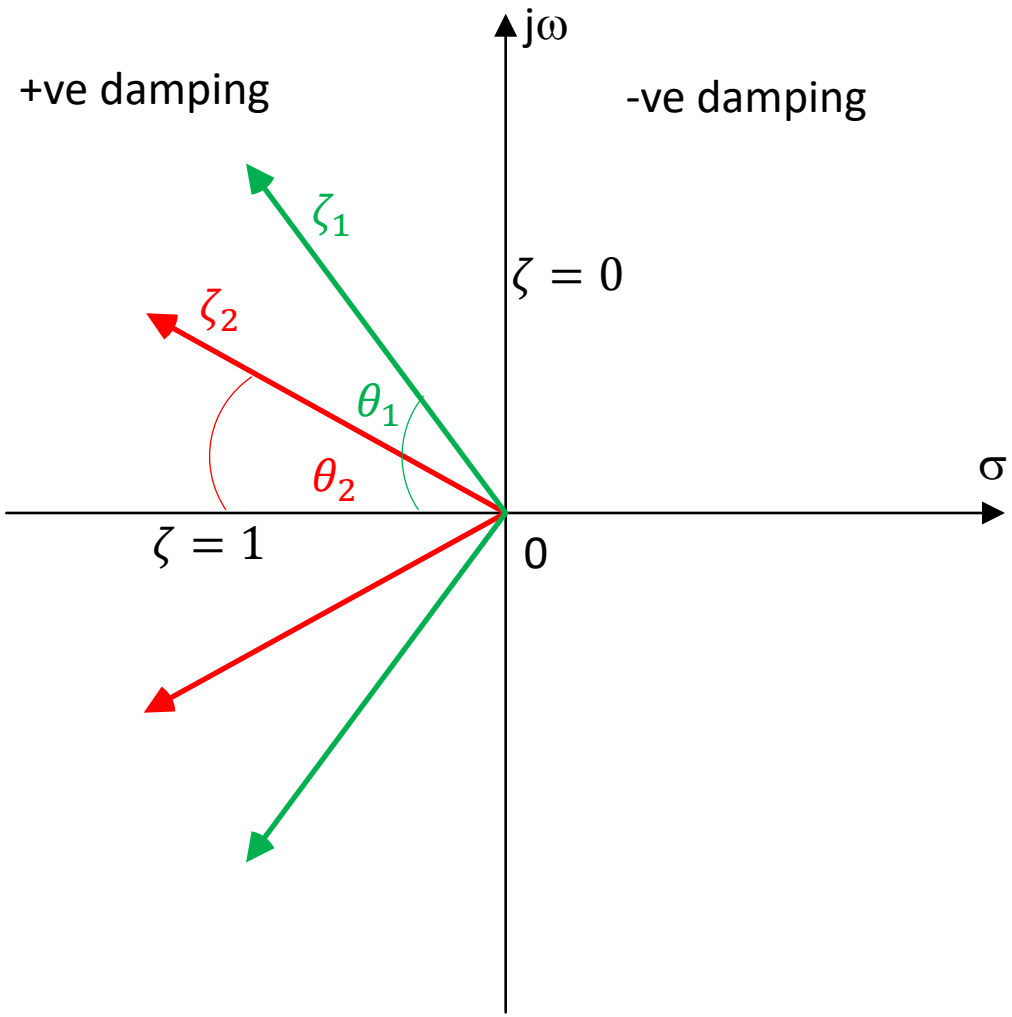
$\cos\theta = \zeta$

2nd order system parameter representation contd.,

Constant ω_n Loci

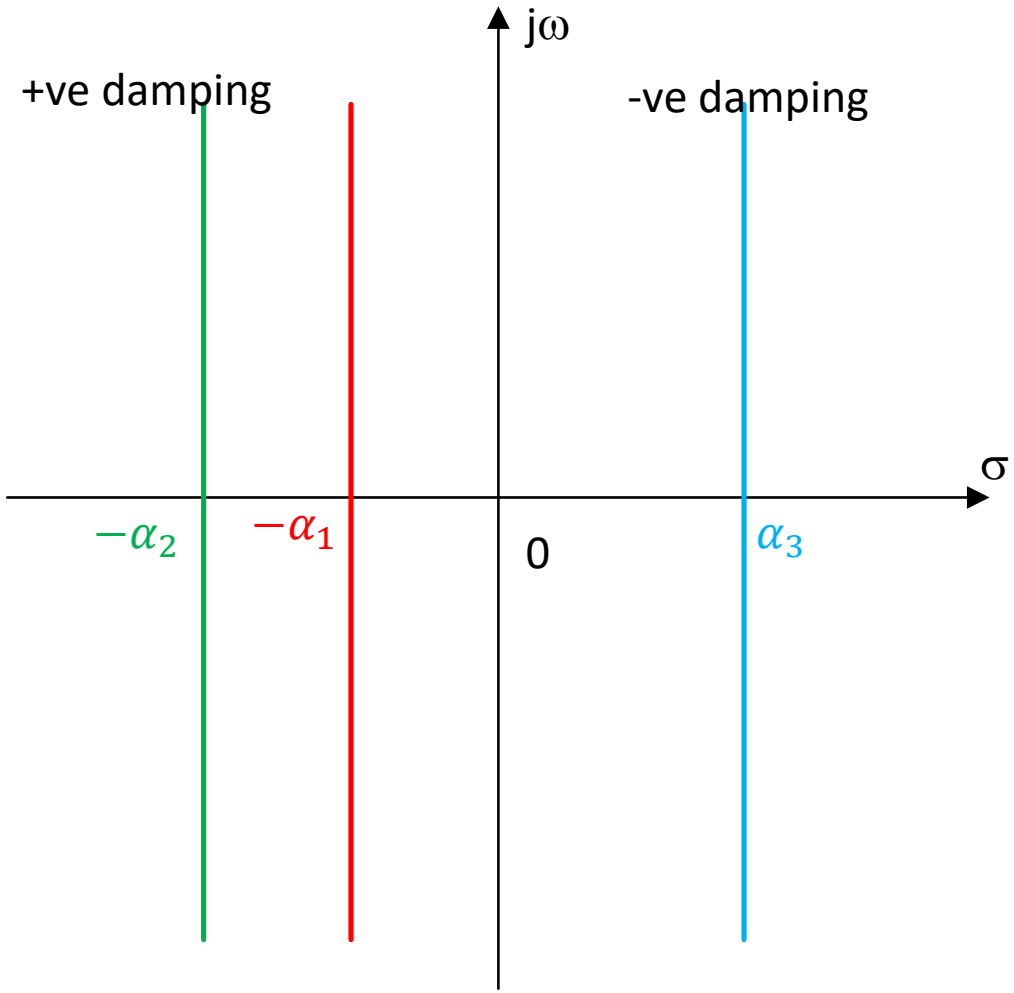


Constant ζ Loci

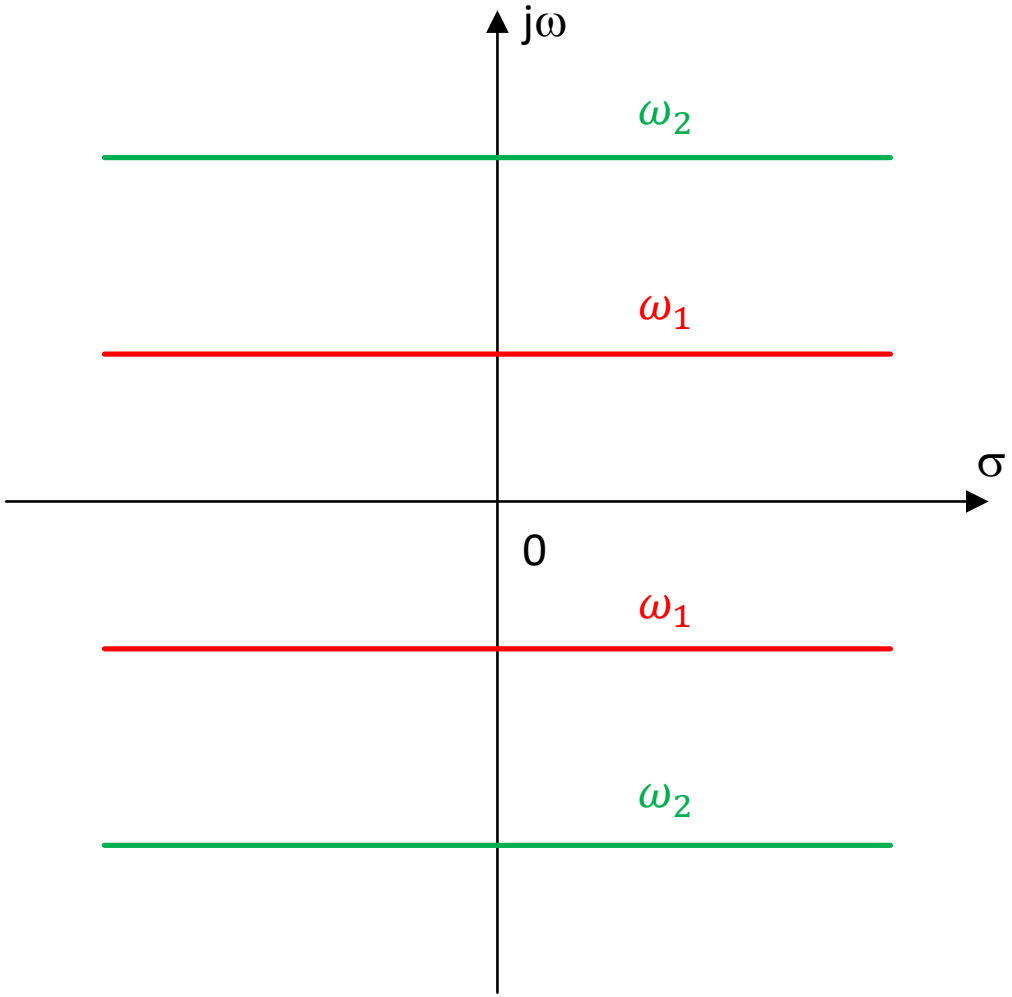


2nd order system parameter representation contd.,

Constant α Loci



Constant ω Loci



Maximum Overshoot

- Since $y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin\left(\omega_n\sqrt{1-\zeta^2}t + \cos^{-1}(\zeta)\right)$
- By differentiating

$$\frac{dy(t)}{dt} = \frac{\omega_n e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\left[\zeta \sin\left(\omega_n\sqrt{1-\zeta^2}t + \cos^{-1}(\zeta)\right) - \sqrt{1-\zeta^2}\cos\left(\omega_n\sqrt{1-\zeta^2}t + \cos^{-1}(\zeta)\right)\right]$$

- With $\cos\theta = \zeta$ and $\omega = \omega_n\sqrt{1-\zeta^2}$

$$\frac{dy(t)}{dt} = \frac{\omega_n e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin\left(\omega_n\sqrt{1-\zeta^2}t\right)$$

- At maximum overshoot $\frac{dy(t)}{dt} = 0$. For that $\omega_n\sqrt{1-\zeta^2}t = n\pi, n = 0,1,2..$

Maximum Overshoot contd.,

- There is another solution for $\frac{dy(t)}{dt} = 0$, that is $t = \infty$. It can happen when $\zeta \geq 1$.
- For unit step response where $0 < \zeta < 1$ the first overshoot is the maximum overshoot, which corresponds to $n = 1$.
- Time for that $t_{max} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$. Corresponding maximum/minimum $y(t)$:
- $y(t)|_{max/min} = 1 - \frac{e^{\left(-n\pi\zeta / \sqrt{1-\zeta^2}\right)}}{\sqrt{1-\zeta^2}} \sin(n\pi + \theta) = 1 + (-1)^{n-1} e^{\left(-n\pi\zeta / \sqrt{1-\zeta^2}\right)}$
- Maximum overshoot is obtained by setting $n = 1$ and obtaining

$$y(t)|_{max/min} - 1 = e^{\left(-\pi\zeta / \sqrt{1-\zeta^2}\right)}$$

Maximum Overshoot contd.,

- $y(t)|_{max/min} - 1 = e^{\left(-\pi\zeta / \sqrt{1-\zeta^2}\right)}$
- Percentage overshoot $100e^{\left(-\pi\zeta / \sqrt{1-\zeta^2}\right)}$

Delay time, Rise time and Settling time

- Delay time (t_d) corresponds to $y(t_d) = 0.5$ which is approximately

$$t_d \approx \frac{1 + 0.7\zeta}{\omega_n}$$

- Rise time (t_r) – time taken to reach from 10% to 90% of the final value, i.e., $t_{10} \rightarrow y(t_{10}) = 0.1, t_{90} \rightarrow y(t_{90}) = 0.9, t_r = t_{90} - t_{10}$

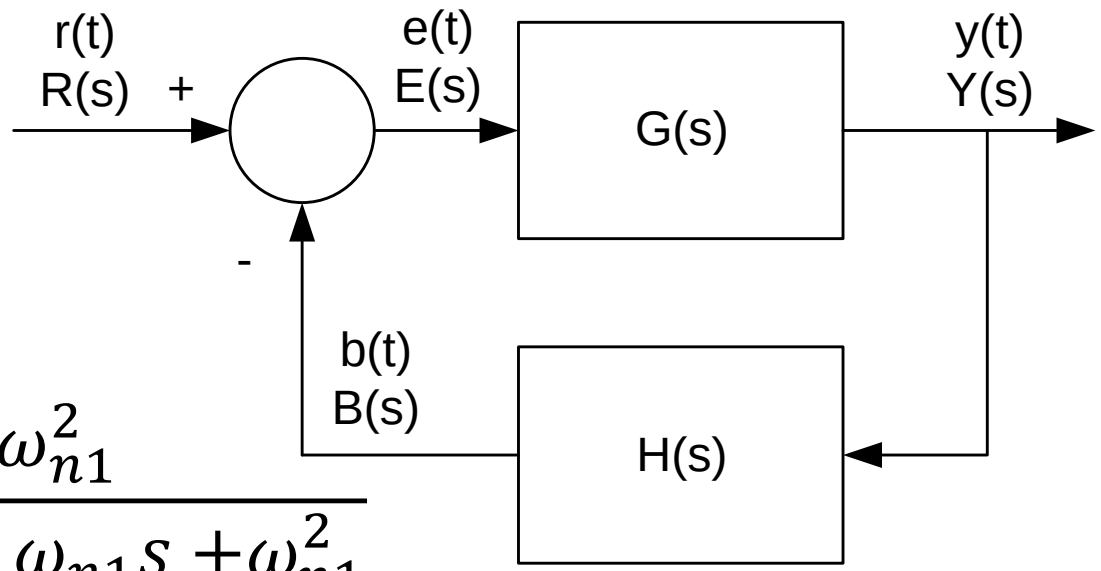
$$t_r \approx \frac{0.8+2.5\zeta}{\omega_n} \text{ for } 0 < \zeta < 1$$

- Settling time (t_s)

$$t_s \approx \frac{3.2}{\zeta\omega_n} \text{ for } 0 < \zeta < 0.69$$

$$t_s \approx \frac{4.5\zeta}{\omega_n} \text{ for } 0.69 < \zeta$$

Design example



- Let $G(s) = \frac{Y(s)}{E(s)} = \frac{\omega_{n1}^2}{s(s+2\zeta_1\omega_{n1})}, H(s) = 1$
- The closed loop system is

$$\frac{Y(s)}{R(s)} = \frac{\omega_{n1}^2}{s^2 + 2\zeta_1\omega_{n1}s + \omega_{n1}^2}$$

- If we want to change the closed loop system to $\frac{Y(s)}{R(s)} = \frac{\omega_{n2}^2}{s^2+2\zeta_2\omega_{n2}s+\omega_{n2}^2}$, we can do so by introducing a controller $C(s)$ in cascade with $G(s)$ such that

$$C(s) = \frac{\omega_{n2}^2(s + 2\zeta_1\omega_{n1})}{\omega_{n1}^2(s + 2\zeta_2\omega_{n2})}$$

- Since the pole of the plant at $s = -2\zeta_1\omega_{n1}$ cancels with the zero of the controller at $s = -2\zeta_1\omega_{n1}$, this method is called **Pole-Zero cancellation**.